Selling The Molecular Kinetic Theory

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Objectives: **Grade level**: 6-12

1. To model aspects of the molecular kinetic theory
2. To observe phenomena influenced by molecular kinetic theory
3. To present credible evidence for the existence of molecules
4. To show the effect of pressure on the boiling point of water
5. To demonstrate how the molecular kinetic theory can explain phase change

Materials needed:

1. Methylene blue dye crystals
2. Two, 600mL beakers
3. Hot and cold water
4. Ping pong ball and paddle
5. Syringe, (30cc-50cc)
6. Rubber bands
7. Wire basket
8. Styrofoam balls, (1.5")
9. Fire syringe
10. Hair dryer
11. Signs: (solid, liquid, gas, freeze, boil, evaporate, melt, condense)

Strategy:

You will try to convince your students that molecules move. You will present a variety of activities and demonstrations to help students internalize concepts associated with the molecular kinetic theory.

1. Fill one 600mL beaker with 500mL of cold water. Fill the other with 500mL of hot water. Set the beakers aside so that any motion in the water comes to rest. Now sprinkle equal quantities of methylene blue on the water in both beakers. Twenty or thirty crystal grains is all that is needed. The blue dye will diffuse through the hot water faster than it will in the cold water. Something moves the dye around so it gets spread out evenly throughout both beakers. However the spreading out, mixing, or diffusing is not done at the same rate of speed. Hot water gets the mixing done faster than cold.

2. Take a rubber band and hold it **close to** your upper lip. Quickly pull the rubber band apart and immediately place it against your upper lip. It will feel warm. When you bring the ends back together, and once again hold it against you lip, the rubber band feels cooler.

This is where I use the ping pong paddle and ball. Get the ping pong ball bouncing rythmically between the paddle and the desk top. If you now quickly lower the paddle you will see and hear a rapid increase in the ping pong ball collisions. If you, on the other hand, raise the paddle, the number of collisions will decrease.

I relate to the students that the ping pong ball and paddle are only a **model** of what occurs on a molecular level in matter. If temperature is the average kinetic energy of moving molecules, then the faster the molecules move the warmer the substance gets. Conversely, the slower the molecules move the cooler the substance gets. The rubber band got hotter or cooler depending on the speed of the molecules that made it up. When we stretched out the rubber band we decreased the volume for the molecules, (modeled by lowering the paddle). When we allowed the rubber band to return to its original shape, we increased the volume for the molecules, (modeled by raising the paddle). When the volume of the rubber band was decreased, (stretched out), the molecules moved faster and resulted in the warmer band you felt against your lip. When

the rubber band returned to its original shape, the molecules had more room. This is why they slowed and **Cooled down**.

3. For the next activity you will use the signs you made. On the board draw the following diagram showing change of phase from solid to gas and from gas to solid.

SOLID" " " " " " "**LIQUID**" " " " " " " " " " **GAS**
MELTING BOILING
EVAPORATING

SOLID$ $ $ $ $ $ $ $**LIQUID**$ $ $ $ $ $ $ $ $ $ **GAS**
FREEZING CONDENSING

I like to tell the class that the melting point and the boiling point are like doors you must go through in order to change the state of matter you're dealing with. Likewise, condensing and freezing are the same doorways that must be passed through again but only in the other direction. This helps students relate heat addition and subtraction as the determining factors in phase change and that substances have a one of a kind doorway to pass through.

Let students play with the wire basket, styrene balls and the hair dryer. The modeling of molecular motion and heat should be readily seen. As we heat a solid, that solids molecules begin to move. Let the hair dryer gently jostle the balls. This models the gradual increase in molecular motion through heating.

Continue increasing the rate of motion by more directly aiming the air from the hair dryer on the balls. When you see the balls moving around and about each other, you have reached the liquid level. Remember this occurred because you added heat to the molecules. In our model the heat is the air from the hair dryer but it could have been from a vacuum cleaner or other cool air source. The air is only a model for the heat energy added.

Finally, turn up the hair dryer full force and point it directly at the basket with the styrene balls in it. This is the model for a gas. Molecules are moving about randomly and most energetically.

4. Some of your students might ask you if the molecular kinetic theory is a two way street. That is, if heating causes molecules to move faster, will moving molecules in some mechanical way cause those molecules to heat up? The answer can be seen by demonstrating the fire syringe.

If you can imagine what happens in a bicycle pump when you compress the handle, you have a fairly good idea about how a fire syringe works. Think about the rhythmic up and down motion of the ping pong ball. As the paddle was brought down towards the table, the ball sped up and ricochetted off the paddle and table much more rapidly. This was the model for increased heat since more molecular motion means more heat.

In the case of the fire syringe, (which is really a lot like a bicycle pump with no opening for air to escape), we smash a gas so quickly that the molecules bang off each other so fast that the temperature quickly rises to at least 451 degrees Fahrenheit....the kindling temperature for paper.

The fire syringe is a good example for students to see that the molecular kinetic theory is indeed reversible.

Performance Assessment:

You are invited to an M. C. Hammer concert. During the concert Hammer says

he is going to test a new men's cologne called "Hammer". He has secretly hidden the cologne under one section of seats and in order to see how strong it is will perform the following test at the concert:

1. The cologne is opened in one section of the stands.
2. As soon as you smell the cologne raise your hand.
3. Time how long the cologne odor takes to travel to all concert goers.

Your challenge is to draw a diagram to show how the concert goers will raise their hands. In addition to the diagram, tell me how the molecular kinetic theory causes the molecules to move.

How will the results of Hammer's test be affected by the temperature at the concert? Will it make a difference if he has an indoor or outdoor concert?